



Online Technical Assessment

Thank you for your interest in the position of Technology Analyst/Intern at Deloitte in Malaysia. We are pleased to invite you to participate in online technical assessment.

This is the first stage of our recruitment process. Please take note that this online technical assessment shall be complete within 2 hours. Please proceed with the questions below, you are required to complete all questions.

1

Full name *

Muhammad Hakim Adlin bin Mohamad Nor

2

Your best email address *

hakimadlin72@gmail.com

HTML Assessment

Please answer all questions in HTML format.

3

Write JavaScript to compare two number, biggest value will prompt "BIG" and smallest value will prompt "SMALL"? *

```
function compareNumbers(a, b) {  
  if (a > b) {
```

```

    alert("BIG");
  } else if (a < b) {
    alert("SMALL");
  } else {
    alert("EQUAL");
  }
}

```

// Example usage:

```

compareNumbers(10, 5); // This will prompt "BIG"
compareNumbers(5, 10); // This will prompt "SMALL"
compareNumbers(7, 7); // This will prompt "EQUAL"

```

4

Write an HTML table tag sequence that outputs the following:

Apple pcs 500 **Apple**
 50 **pcs** 5 **Apple** *

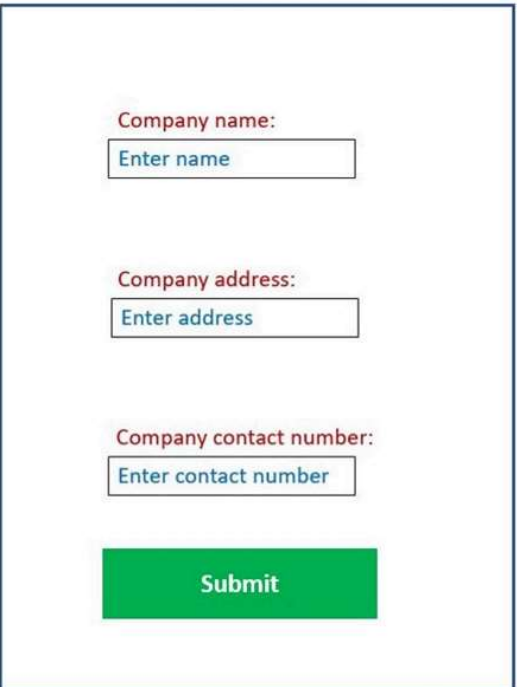
```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Table Example</title>
  <style>
    .bold {
      font-weight: bold;
    }
  </style>
</head>
<body>
  <table border="1">
    <tr>
      <td class="bold">Apple pcs</td>
      <td>500</td>
      <td class="bold">Apple</td>
    </tr>
    <tr>
      <td>50</td>
      <td class="bold">pcs</td>
      <td>5</td>
      <td class="bold">Apple</td>
    </tr>
  </table>
</body>
</html>

```

5

Write a HTML tag that will output the following form: *



The form is enclosed in a blue border. It contains three input fields, each with a red label above it. The first input field is labeled 'Company name:' and contains the placeholder text 'Enter name'. The second input field is labeled 'Company address:' and contains the placeholder text 'Enter address'. The third input field is labeled 'Company contact number:' and contains the placeholder text 'Enter contact number'. Below the input fields is a green 'Submit' button.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Company Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }
    .form-container {
      border: 1px solid #000;
      padding: 20px;
      width: 300px;
    }
    .form-group {
      margin-bottom: 15px;
    }
    .form-group label {
      display: block;
      color: red;
      margin-bottom: 5px;
    }
    .form-group input {
      width: 100%;
      padding: 8px;
      box-sizing: border-box;
    }
  </style>
</head>
<body>
  <div class="form-container">
    <div class="form-group">
      <label>Company name:</label>
      <input type="text" value="Enter name">
    </div>
    <div class="form-group">
      <label>Company address:</label>
      <input type="text" value="Enter address">
    </div>
    <div class="form-group">
      <label>Company contact number:</label>
      <input type="text" value="Enter contact number">
    </div>
    <div>Submit</div>
  </div>
</body>
</html>
```

```

        .submit-btn {
            width: 100%;
            padding: 10px;
            background-color: green;
            color: white;
            border: none;
            cursor: pointer;
        }
    </style>
</head>
<body>
    <div class="form-container">
        <form>
            <div class="form-group">
                <label for="company-name">Company name:</label>
                <input type="text" id="company-name" name="company-name" placeholder="Enter name">
            </div>
            <div class="form-group">
                <label for="company-address">Company address:</label>
                <input type="text" id="company-address" name="company-address" placeholder="Enter address">
            </div>
            <div class="form-group">
                <label for="company-contact">Company contact number:</label>
                <input type="text" id="company-contact" name="company-contact" placeholder="Enter contact number">
            </div>
            <button type="submit" class="submit-btn">Submit</button>
        </form>
    </div>
</body>
</html>

```

Programming

6

Describe the differences between SOAP and REST? *

SOAP (Simple Object Access Protocol) is a protocol with strict rules that relies on XML-based messaging to ensure platform and language independence, featuring built-in error handling and security mechanisms like WS-Security. It is typically used in complex, distributed enterprise environments with tightly coupled client-server interactions and service interfaces described by WSDL. In contrast, REST (Representational State Transfer) is an architectural style that uses standard HTTP methods (GET, POST, PUT, DELETE) for communication and supports multiple data formats, including JSON and XML. RESTful services are loosely coupled, follow a resource-oriented approach, and are stateless, making them easier to scale horizontally and more suitable for high-performance web applications. While SOAP's extensive XML usage can lead to higher overhead and slower performance, REST is generally faster and more efficient. Security in REST is typically implemented via HTTPS and other mechanisms like OAuth and JWT. SOAP is favored in scenarios requiring high security, reliability, and transaction compliance, whereas REST is ideal for lightweight, scalable, and flexible web services, often used in web and mobile applications.

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What are some important annotations which you use to create RESTful web services? *

@Path, @GET, @POST, @PUT, @DELETE, @PathParam, @QueryParam

8

Complete the Java function below to print 2 biggest number from the array. *

JAVA:

```
public static void main(String[] args)
{
    int num[] = {5,34,78,2,45,1,98,23,97,-45};
    printTwoBiggestNumbers(num);
}

public void printTwoBiggestNumbers(int[] n)
{
    //complete your program here
}
```

Expected Output:

```
First Biggest Number: 98
Second Biggest Number: 97
```

```
public static void main(String[] args) {

    int[] num = {5, 34, 78, 2, 45, 1, 98, 23, 97, -45};
    printTwoBiggestNumbers(num);

}

public static void printTwoBiggestNumbers(int[] n) {

    int firstBiggest = Integer.MIN_VALUE;
    int secondBiggest = Integer.MIN_VALUE;

    for (int i = 0; i < n.length; i++) {

        if (n[i] > firstBiggest) {
            secondBiggest = firstBiggest;
            firstBiggest = n[i];
        }
    }
}
```

```

    } else if (n[i] > secondBiggest && n[i] != firstBiggest) {
        secondBiggest = n[i];
    }
}

System.out.println("First Biggest Number: " + firstBiggest);
System.out.println("Second Biggest Number: " + secondBiggest);
}

```

Programming

9

Write a Java program which loop numbers from 1 to 100. For multiples of 7 print "Ta" instead of the number and print "Da" for the multiples of 9. When number is divided by 10, print "TaDa". *

Expected Output:-

```

7: Ta
9: Da
10: TaDa
14: Ta
18: Da
20: TaDa
21: Ta
27: Da
28: Ta
30: TaDa

```

```

public class TaDaPrinter {
    public static void main(String[] args) {
        for (int i = 1; i <= 100; i++) {
            if (i % 10 == 0) {
                System.out.println("TaDa");
            } else if (i % 7 == 0) {
                System.out.println("Ta");
            } else if (i % 9 == 0) {
                System.out.println("Da");
            } else {
                System.out.println(i);
            }
        }
    }
}

```

10

Complete the Java function below to removing the duplicates from the array.

*

JAVA:

```
public static void main(String[] args)
{
    System.out.print("The value is " +
removeDuplicate(1,2,3,4,4,4,4,4,5,5,5,6,5,5,5,5,5,8));
}

public static String removeDuplicate (int[] p)
{
    //complete your program here
}
```

Expected Output:-

```
The value is 1,2,3,4,5,6,8
```

```
import java.util.LinkedHashSet;
import java.util.Set;
```

```
public class Main {
    public static void main(String[] args) {
        System.out.print("The value is " + removeDuplicate(new int[]{1, 2, 3, 4, 4, 4, 4, 5, 5, 6, 5, 5, 5, 8}));
    }

    public static String removeDuplicate(int[] p) {
        Set<Integer> set = new LinkedHashSet<>();
        for (int num : p) {
            set.add(num);
        }

        StringBuilder result = new StringBuilder();
        for (int num : set) {
            if (result.length() > 0) {
                result.append(",");
            }
            result.append(num);
        }

        return result.toString();
    }
}
```

Programming

11

Write a program that is able to calculate the total amount while giving a discount which can be up to a maximum of 15 % based on the quantity of fruits. Each increment of 12 in fruits will increase discount by 1%. *

JAVA:

```
public static void main(String[] args){

    String[][] itemArr = {"Durian", "45.50", "50"},
                        {"Apple", "5.50", "6"},
                        {"Orange", "6.00", "10"};

    calculateAndPrint(itemArr);

}

private static void calculateAndPrint(String[][] itemArr){
    //Complete your program here
}
```

Expected Output:

```
You have bought 50 Durian with total of RM 2275.0
You have bought 6 Apple with total of RM 33.0
You have bought 10 Orange with total of RM 60.0
Total discount rate: 5%
Total amount paid: RM 2249.6
```

```
public class Main {
    public static void main(String[] args) {
        String[][] itemArr = {
            {"Durian", "45.50", "50"},
            {"Apple", "5.50", "6"},
            {"Orange", "6.00", "10"}
        };

        calculateAndPrint(itemArr);
    }

    private static void calculateAndPrint(String[][] itemArr) {
        double totalAmount = 0.0;
        int totalQuantity = 0;

        for (String[] item : itemArr) {
            String name = item[0];
            double price = Double.parseDouble(item[1]);
            int quantity = Integer.parseInt(item[2]);
            double itemTotal = price * quantity;
```

```

totalAmount += itemTotal;
totalQuantity += quantity;

System.out.printf("You have bought %d %s with total of RM %.1f%n", quantity, name, itemTotal);
}

int discountRate = Math.min(totalQuantity / 12, 15);
double discount = (discountRate / 100.0) * totalAmount;
double totalPaid = totalAmount - discount;

System.out.printf("Total discount rate: %d%%\n", discountRate);
System.out.printf("Total amount paid: RM %.1f%n", totalPaid);
}
}

```

SQL

12

Given 2 tables below, write a single SQL query to display below result *

T CUSTOMER

CUST_ID	FIRST_NAME	CITY
CUS001	David	New York
CUS002	Quin	Berlin
CUS003	Lalisa	Moscow
CUS004	Kaelyn	Paris
CUS005	James	Berlin

T ORDER

ORDER_NO	ORDER_AMT	ORDER_DATE	CUST_ID
ORD0001	558.00	01-02-2020	CUS001
ORD0002	345.55	02-02-2020	CUS004
ORD0003	664.80	02-02-2020	CUS003
ORD0004	340.90	06-02-2020	CUS004
ORD0005	262.13	06-02-2020	CUS001
ORD0006	121.20	10-02-2020	CUS002

Write SQL query to display below result:

- Result should ONLY contain 3 columns: **FIRST_NAME, TOTAL_AMT, CITY**
- **TOTAL_AMT** column is sum of order amount of respective customer
- Do not show customers from **CITY = Berlin**
- Order data by **TOTAL_AMT descending**

Expected Output:-

FIRST_NAME	TOTAL_AMT	CITY
David	820.13	New York
Kaelyn	686.45	Paris
Lalisa	664.8	Moscow

```

SELECT C.FIRST_NAME, SUM(O.ORDER_AMT) AS TOTAL_AMT, C.CITY
FROM CUSTOMER C
JOIN `ORDER` O ON C.CUST_ID = O.CUST_ID
WHERE C.CITY != 'Berlin'

```

GROUP BY C.FIRST_NAME, C.CITY
ORDER BY TOTAL_AMT DESC;



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